DxtER (Credit: Bloomberg)

re-admissions, mediclaim processing and image recognition for diagnosis. These can help schedule doctor appointments based on the severity of symptoms, thus minimising staffing challenges.

Blockchain-based solutions. It is easy to detect fraudulent drug dealers with blockchain-maintained transactions. Blockchain ensures authenticity and high quality of registered medicines. It creates transparency by recording operational data when a drug is produced and moved from the manufacturer to retailer. This makes it easy to prove the authenticity of any registered document, verify the whole path of the drug movement and determine all chain links at any time.

Blockchain allows recording, storing and sharing of results from clinical tests in a secure way. It makes it impossible to modify data as it generates a unique hash every time a modification is made to the data. A patient can decide whom to provide access and whether this access will be full or partial. A few blockchain healthcare startups that have made this technology the base of their operational structure are Guardtime, Gem Health, Cyph, MedRec and Blockchain Health.

Electronics to transform the medical industry

Complete diagnostic tool. DxtER by Basil Leaf is a diagnostic tool that can identify 34 illnesses, including pneumonia, diabetes and atrial fibrillation. It uses a suite of Bluetooth sensors to connect to an app on the tablet. The app informs about the health status through sensors applied on the chest patch, finger probe, spirometer and a combination of thermometer-stethoscope using automated dialogue. Basil Leaf is working on DxtER upgrades that will enable it to use the same hardware to diagnose as many as 75 conditions, including respiratory illnesses and a wider range of cardiac ailments.

Neuromodulation to treat obstructive sleep apnea. Sleep apnea can lead to high blood pressure, heart diseases and stroke. A continuous positive airway pressure device used for treatment is an implant that delivers stimulation to open key airway muscles during sleep. It can be controlled by a remote or wearable patch, the technology acting as a pacemaker. It helps synchronise the intake of air with the action of the tongue with the help of a breathing sensor and a stimulation lead powered by a small battery.

Smart inhaler (Credit: Dr Neil Paul's Blog)

Centralised monitoring of hospital patients. This includes sensors and high-definition cameras around a patient to monitor blood pressure, respiration, heart rate and more. An alert is generated for doctors or on-site personnel that help is needed.

Smart inhalers. A Bluetooth-enabled smart inhaler has been designed to detect need for inhaler use, remind patients to take medicine, encourage proper use of the device and gather data about inhaler use that can help take proper care. A smart inhaler is effective when it is used correctly, so each time it is used, it records the date, time, place and whether the dose was correctly administered. It can also send data to a connected smartphone app.

Propeller Health has designed a GPS-enabled inhaler that has a Bluetooth transmitter to communicate with an asthma patient's caretaker's smartphone. This sensor activates when the inhaler is pressed and records the exact time and location of a patient's asthma attack on a smartphone app.

Organs on a chip. Wyss Institute has created organs on a chip by making advancements in DNA sequencing and through in-depth stem cell research. These are electronic sensors that are used to test different kinds of treatment possibilities to choose the best, and apply the same to the patient. These chips mimic different human organs like lungs, kidney, heart, gut, bone marrow and so on. Cells on the chip grow and respond exactly like real cells found in human organs. Now, drugs can be tested safely before being tried on humans. This technology will also prevent killing of several animals due to medical tests being conducted on them.

Artificial pancreas. This is a fully-automated, closed-